Statistics (2)

Syllabus

Yu-You Liou

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Lecturer: Yu-You Liou	**Contact**: d10627008@ntu.edu.tw
Course No.: EIB-21F-01-A1	**Website**: https://yyliou.github.io/stat
Meeting Time: Thursday, Periods 6–7 (13:10–15:00)	**Location**: L309

Course Objectives

1. To train students in basic statistical knowledge and econometric concepts.
2. To enhance students' familiarity with data analysis and provide hands-on experience in statistical analysis.

Textbooks

1. Bluman, A. G. (2012). *Elementary Statistics: A Step-by-Step Approach* (8th ed.). McGraw-Hill.

Course Structure

1. **Lecture** (First hour): Foundational knowledge and concepts introduced each session.
2. **Lab** (Second hour): Hands-on in-class exercises using R to apply the concepts learned during the lecture.

Assessment Criteria

1. **Attendance** (12%): $R = \min\{12, t\}$, where t represents the number of sessions attended.
2. **In-Class Exercises** (48%): 12 exercises $\times$ 4% each, submitted at the end of each session.
3. **Mid-Term Examination** (20%) and **Final Examination** (20%).

Regulations

1. **Attendance**

Per Shih Chien University regulations, students who are absent for more than one-third of class sessions will receive a final grade of zero. Make-up attendance will not be permitted.

2. **Examination**

Students who do not show up for their scheduled examination will receive a grade of zero for that component.

3. **In-class exercise deadline**

The deadline for all in-class exercises is before midnight of the following week. Late submissions will not be accepted.

4. **Academic Integrity**

All submitted work must be the student's own. Copying from other students is strictly prohibited. AI detection tools are used to review submissions; any work found to be copied will receive a grade of zero.

5. **AI Policy**

Students may use AI tools (e.g., ChatGPT, GitHub Copilot) as learning aids, but must be able to explain and defend any code or analysis they submit. Uncritical submission of AI-generated work without understanding constitutes academic dishonesty.

Required Software

All software is free and open-source: **R** at https://cran.r-project.org, **RStudio Desktop** at https://posit.co/download/rstudio-desktop, and R packages `tidyverse`, `ggplot2`, `patchwork` (installation instructions on the course R page).

Course Schedule

Week	Topic
1	Explain the syllabus and class regulations
2	**Chapter 8**: Hypothesis testing (ex1)
3	**Chapter 8**: Hypothesis testing (ex2)
4	**Chapter 8**: Hypothesis testing (ex3)
5	**Chapter 9**: Testing the difference between two means, two proportions, and two variances (ex4)
6	**Chapter 9**: Testing the difference between two means, two proportions, and two variances (ex5)
7	**Chapter 10**: Correlation and regressions (ex6)
8	Midterm review
9	Midterm exam
10	**Chapter 10**: Correlation and regression (ex7)
11	**Chapter 10**: Correlation and regression (ex8)
12	**Chapter 11**: Other chi-square tests (ex9)
13	**Chapter 11**: Other chi-square tests (ex10)
14	**Chapter 12**: Analysis of variance (ex11)
15	**Chapter 12**: Analysis of variance (ex12)
16	Final exam
17	Flexible learning week
18	Flexible learning week

Note: Syllabus is subject to change. Any updates will be announced on the course website.